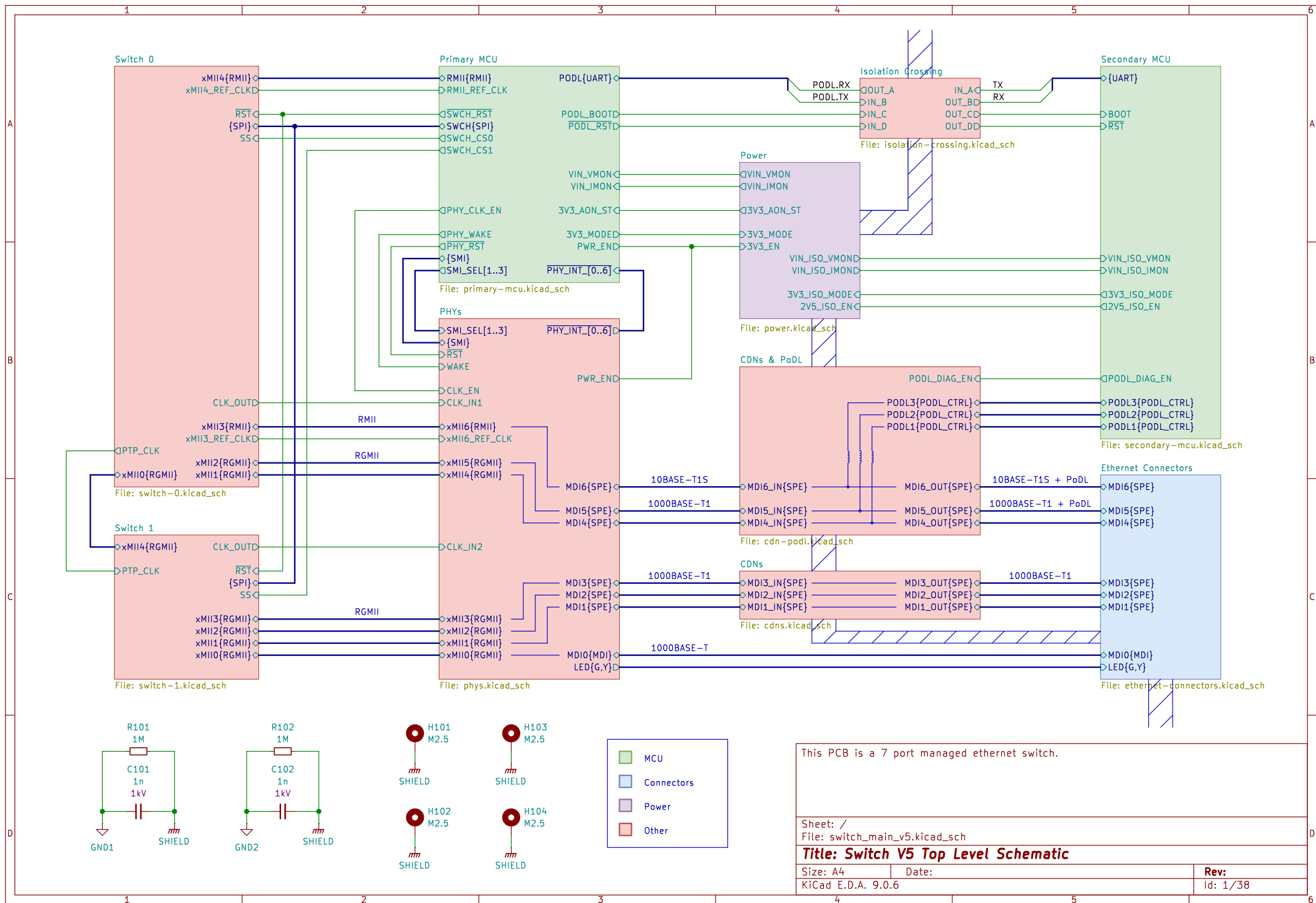
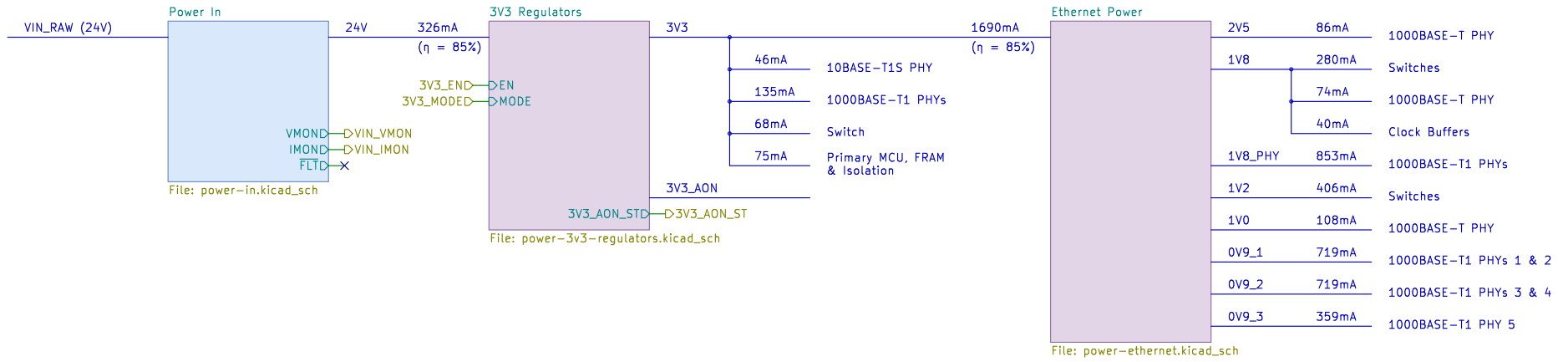


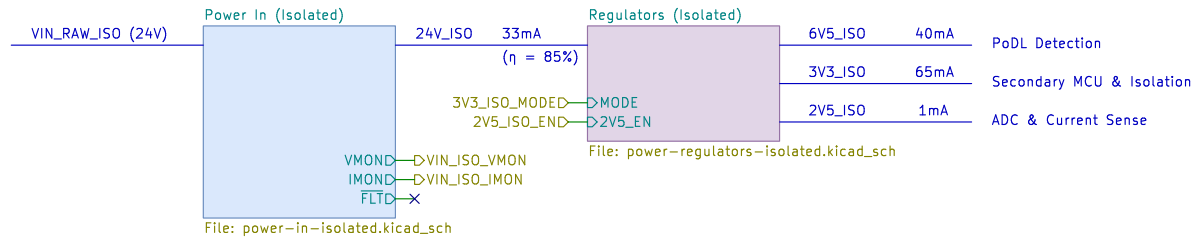


PHY Side Power



Isolation Barrier

Connector Side Power



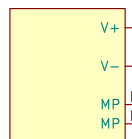
Block diagram of the power supply scheme for both sides of the isolation barrier.

Sheet: /Power/
File: power.kicad_sch

Title: Power	
Size: A4	Date:
KiCad E.D.A. 9.0.6	

Rev:
Id: 2/38

J301
MKXS1-1 Power Input



ESD
Protection

Surge
Protection

VIN_RAW

D301
SZESD7241N2T5G



U301
TVS3301DRBR

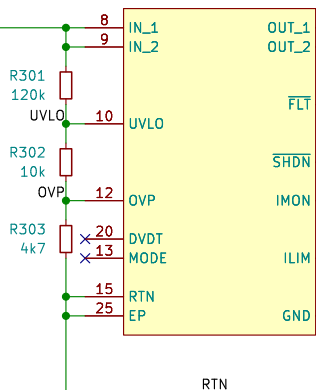
C301
100n
100V

GND1

UVLO = 10.9V
OVP = 34.1V

eFuse

U302
TPS26600RHFR



$VMON = VIN / 18.857$
The 150mΩ eFuse
resistance should be
taken into account when
calculating input power.

+VIN

R307
100k

R308
5k6

GND1

R306
100k

IMON

$I_{OUT} = VMON / 7.84$

$R_{ILIM} = 47k\Omega \therefore$
 $I_{LIM} = 285mA$

TPS26600 has auto-retry
when a fault occurs.

Power input and protection for the PHY side of the PCB.

Sheet: /Power/Power In/
File: power-in.kicad_sch

Title: Power In (PHY Side)

Size: A4

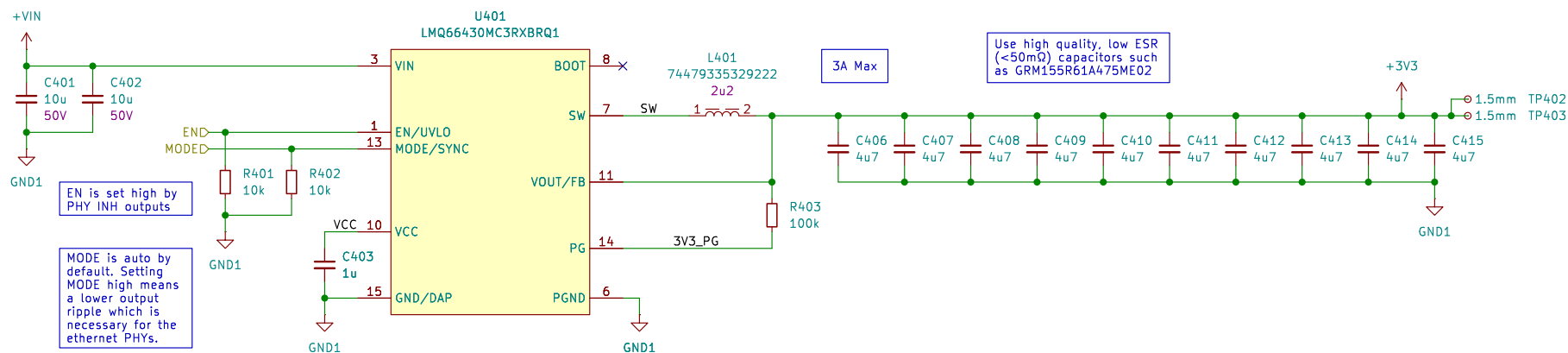
Date:

KiCad E.D.A. 9.0.6

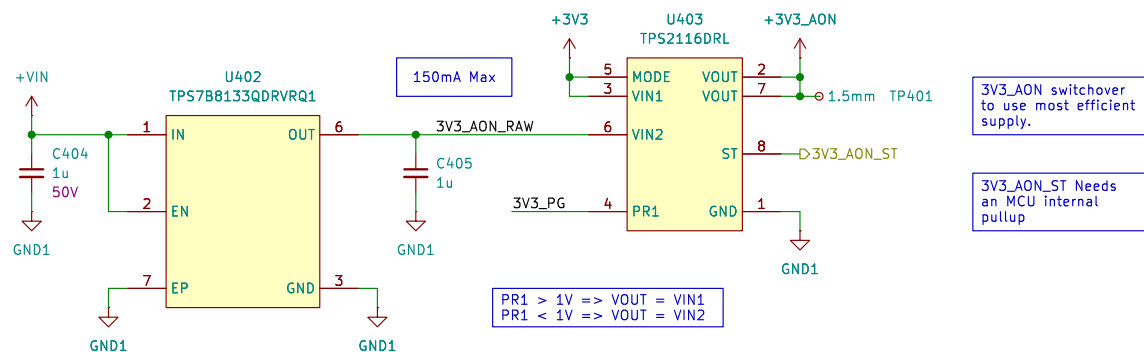
Rev:

Id: 3/38

3V3 Main Regulator



3V3 Always on supply



3V3 and 3V3_AON (always on) voltage regulators.

Sheet: /Power/3V3 Regulators/
File: power-3v3-regulators.kicad_sch

Title: 3V3 Regulators (PHY Side)

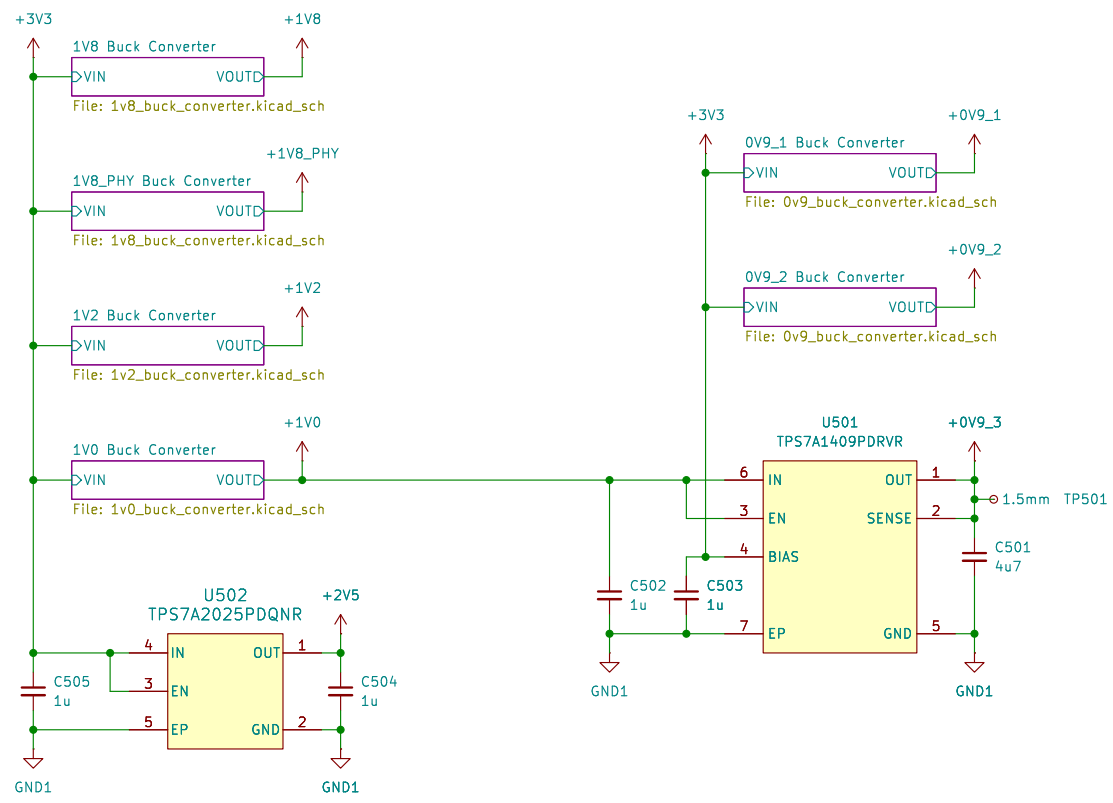
Size: A4

Date:

KiCad E.D.A. 9.0.6

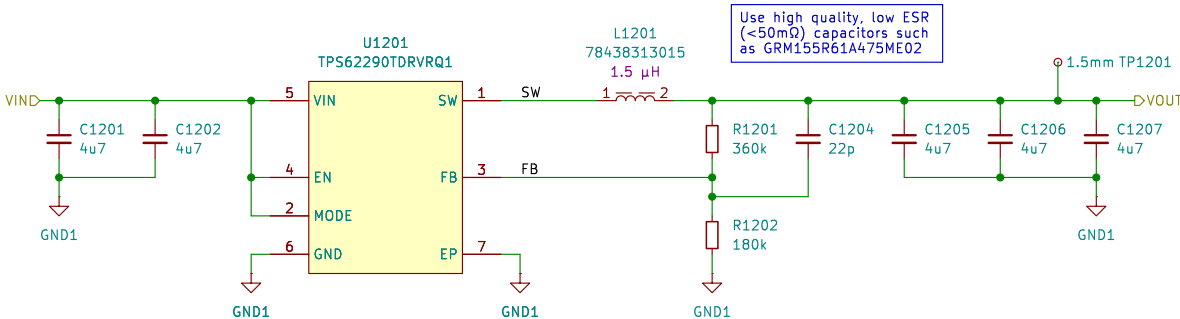
Rev:

Id: 4/38



1V8 and 1V2 SMPS for ethernet switch and PHYs.		
Sheet: /Power/Ethernet Power/ File: power-ethernet.kicad_sch		
Title: Ethernet Power (PHY Side)		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 5/38

1V8 Regulator



Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

Sheet: /Power/Ethernet Power/1V8 Buck Converter,
File: 1v8_buck_converter.kicad_sch

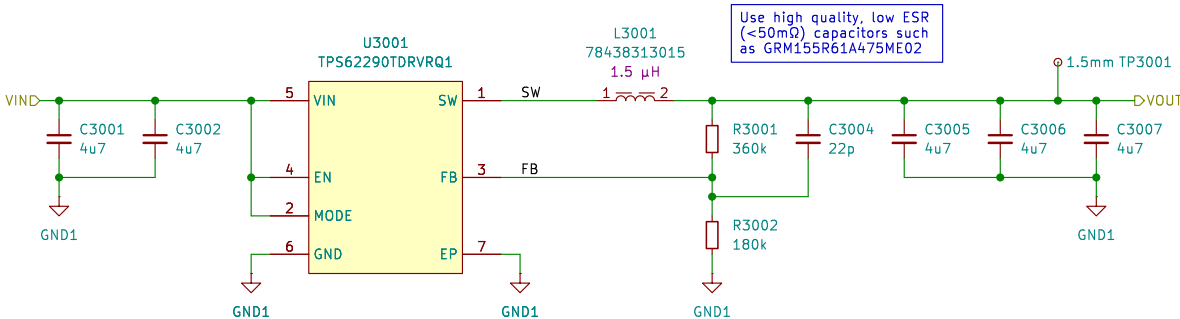
Title:

Size: A4	
KiCad E.D.A. 9.0.6	

Date:

Rev: 12/38

1V8 Regulator



Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

Sheet: /Power/Ethernet Power/1V8_PHY Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

Size: A4

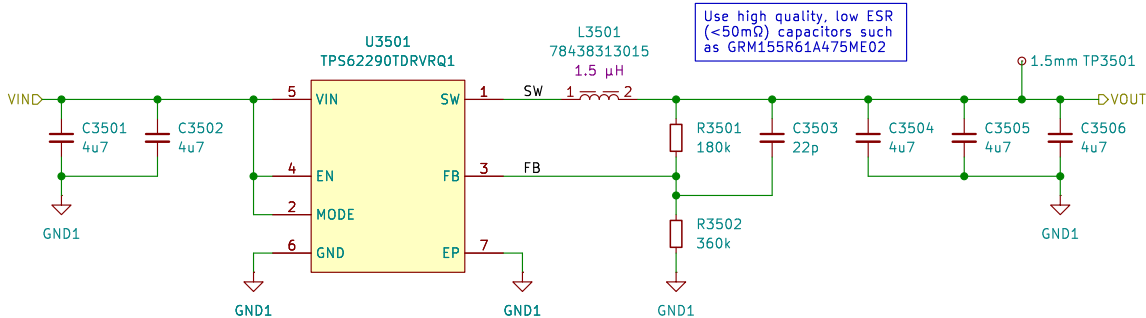
Date _____

KiCad E.D.A. 9.0.6

Rev:

Id: 30/38

0V9 Regulator



Sheet: /Power/Ethernet Power/0V9_1 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

Size: A4

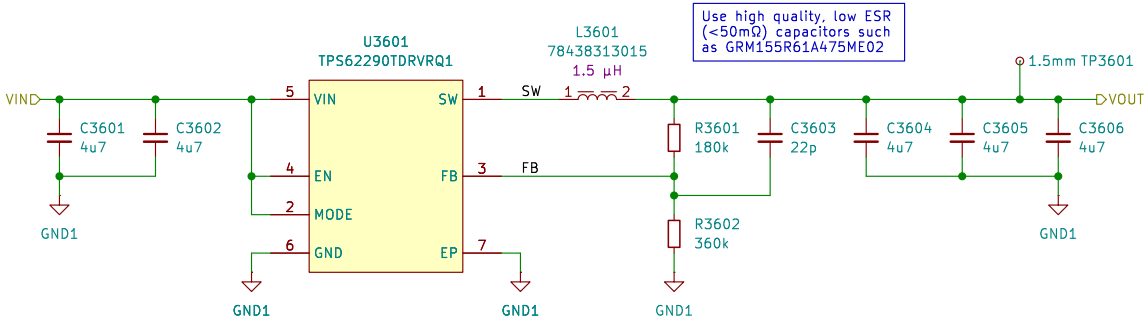
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 35/38

0V9 Regulator



Sheet: /Power/Ethernet Power/0V9_2 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

Size: A4

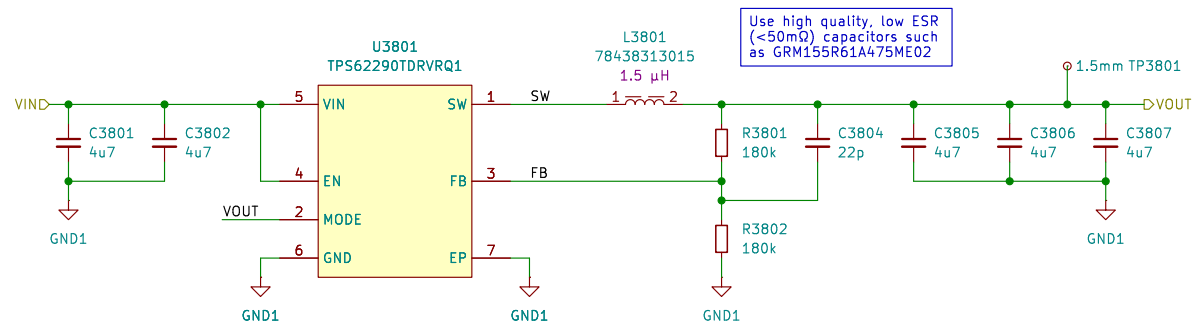
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 36/38

1V2 Regulator



Sheet: /Power/Ethernet Power/1V2 Buck Converter/
File: 1v2_buck_converter.kicad_sch

Title:

Size: A4

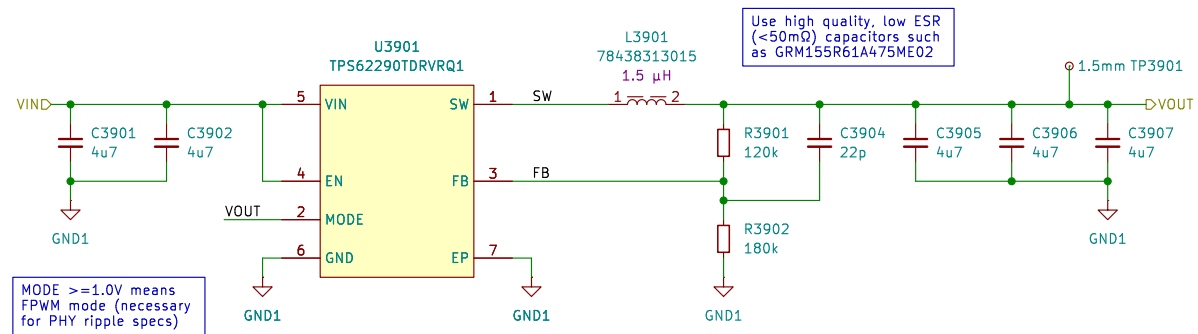
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 38/38

1V0 Regulator



Sheet: /Power/Ethernet Power/1V0 Buck Converter/
File: 1v0_buck_converter.kicad_sch

Title:

Size: A4

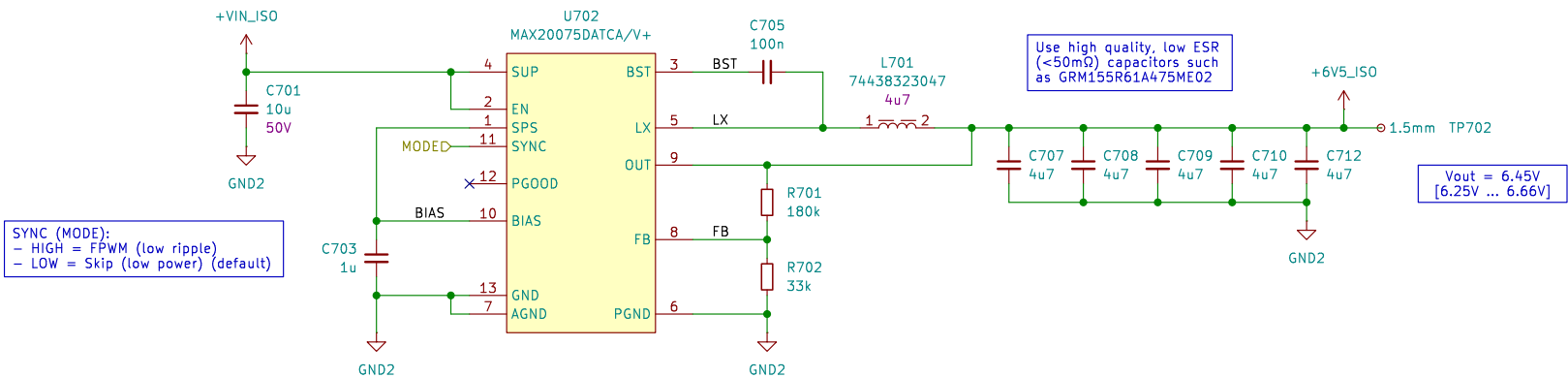
Date:

KiCad E.D.A. 9.0.6

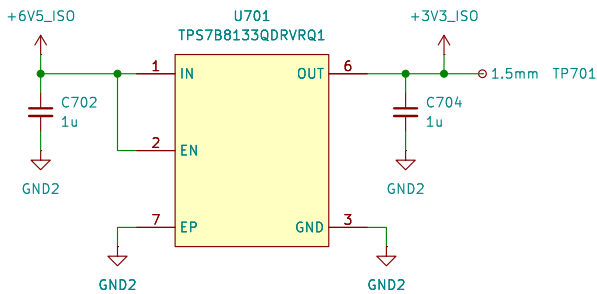
Rev:

Id: 39/38

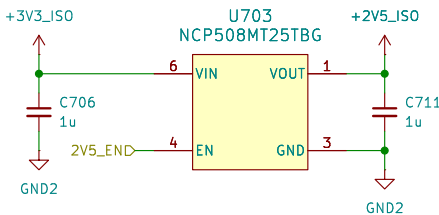
6V5 Regulator



3V3 Regulator



2V5 Regulator



6V5, 3V3 and 2V5 regulators for the connector side of the PCB.

Sheet: /Power/Regulators (Isolated)/
File: power-regulators-isolated.kicad_sch

Title: Regulators (Connector Side)

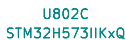
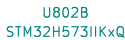
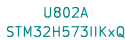
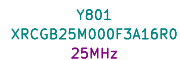
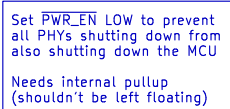
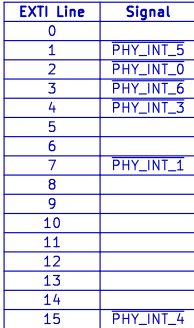
Size: A4

Date:

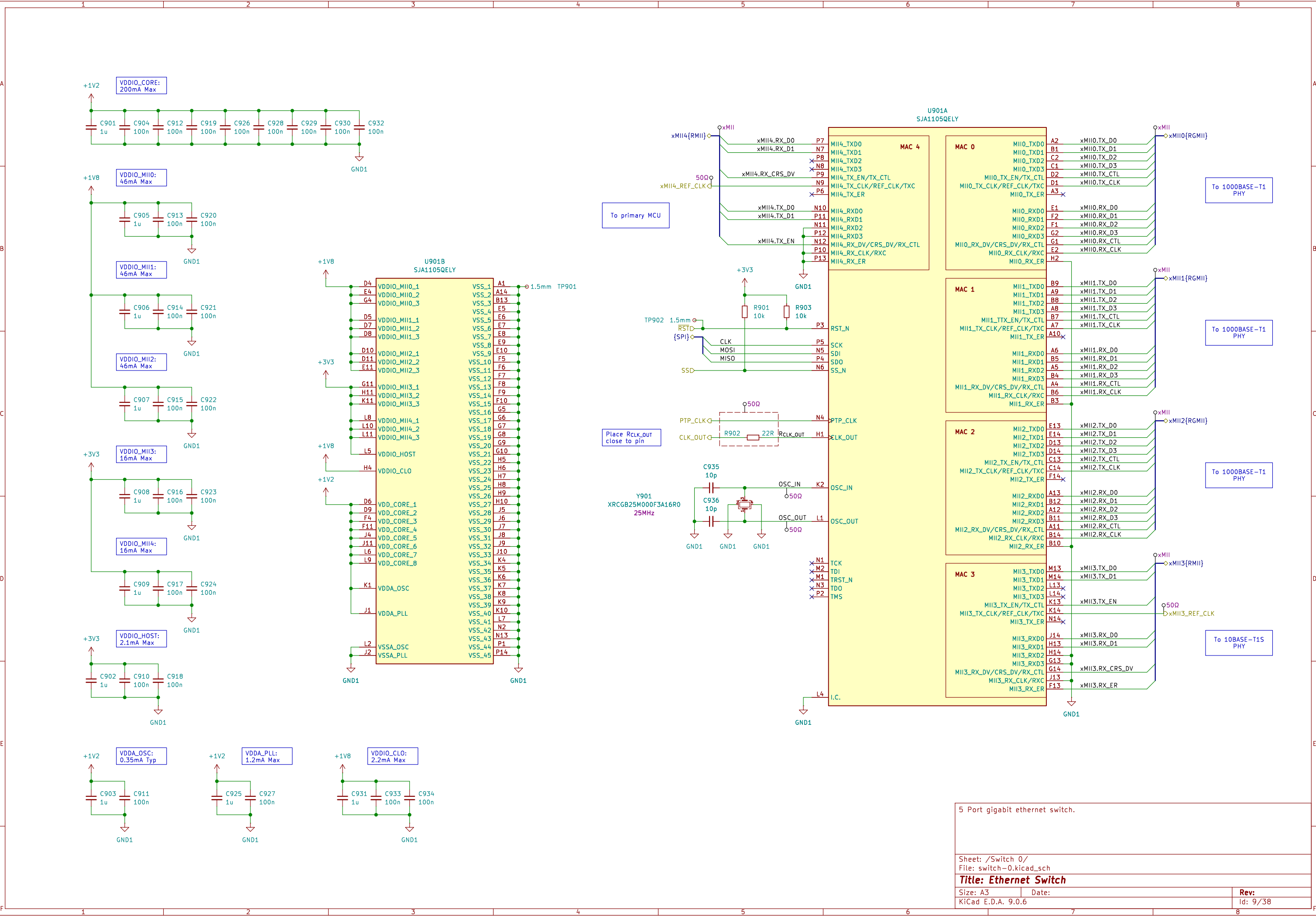
KiCad E.D.A. 9.0.6

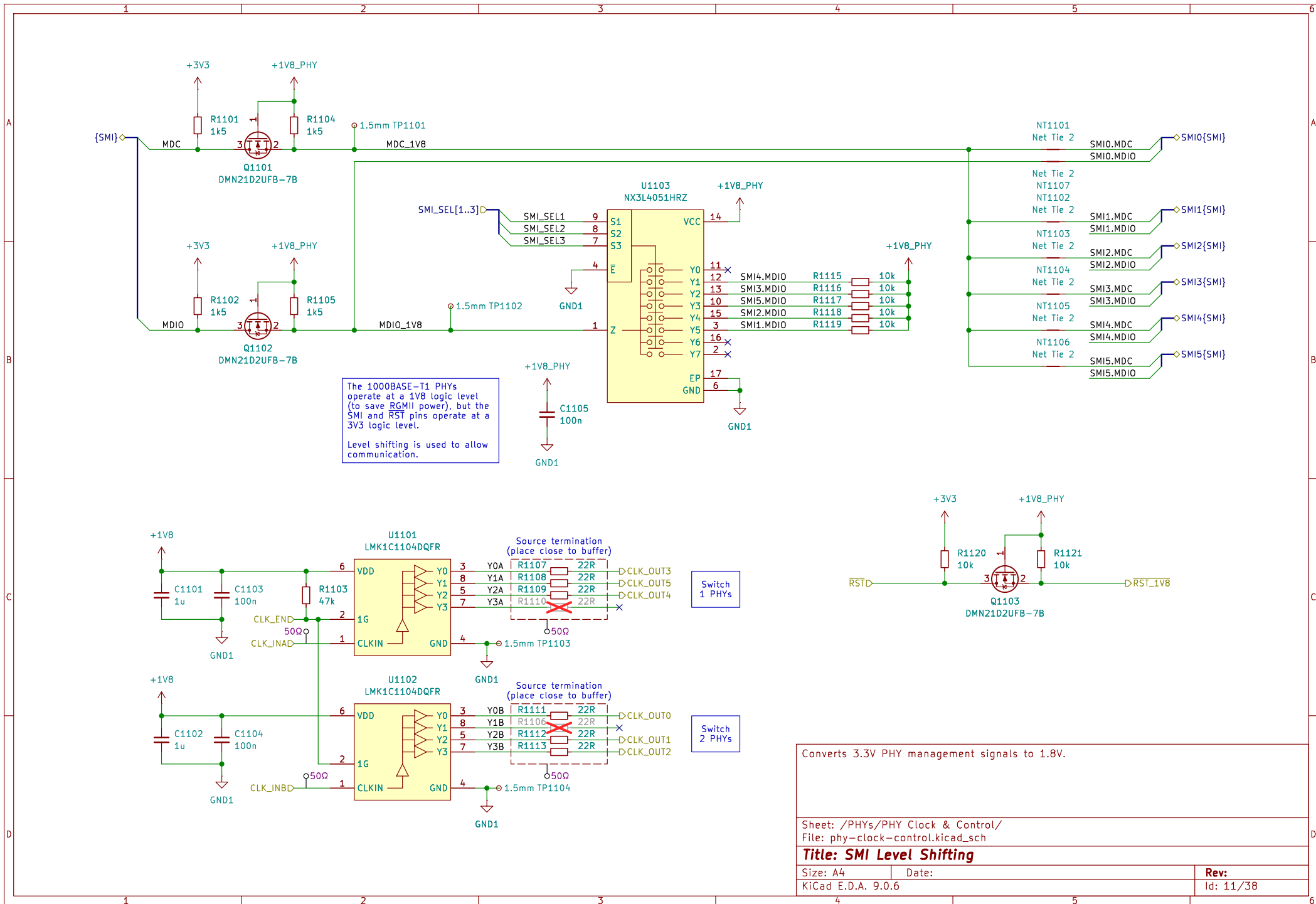
Rev:

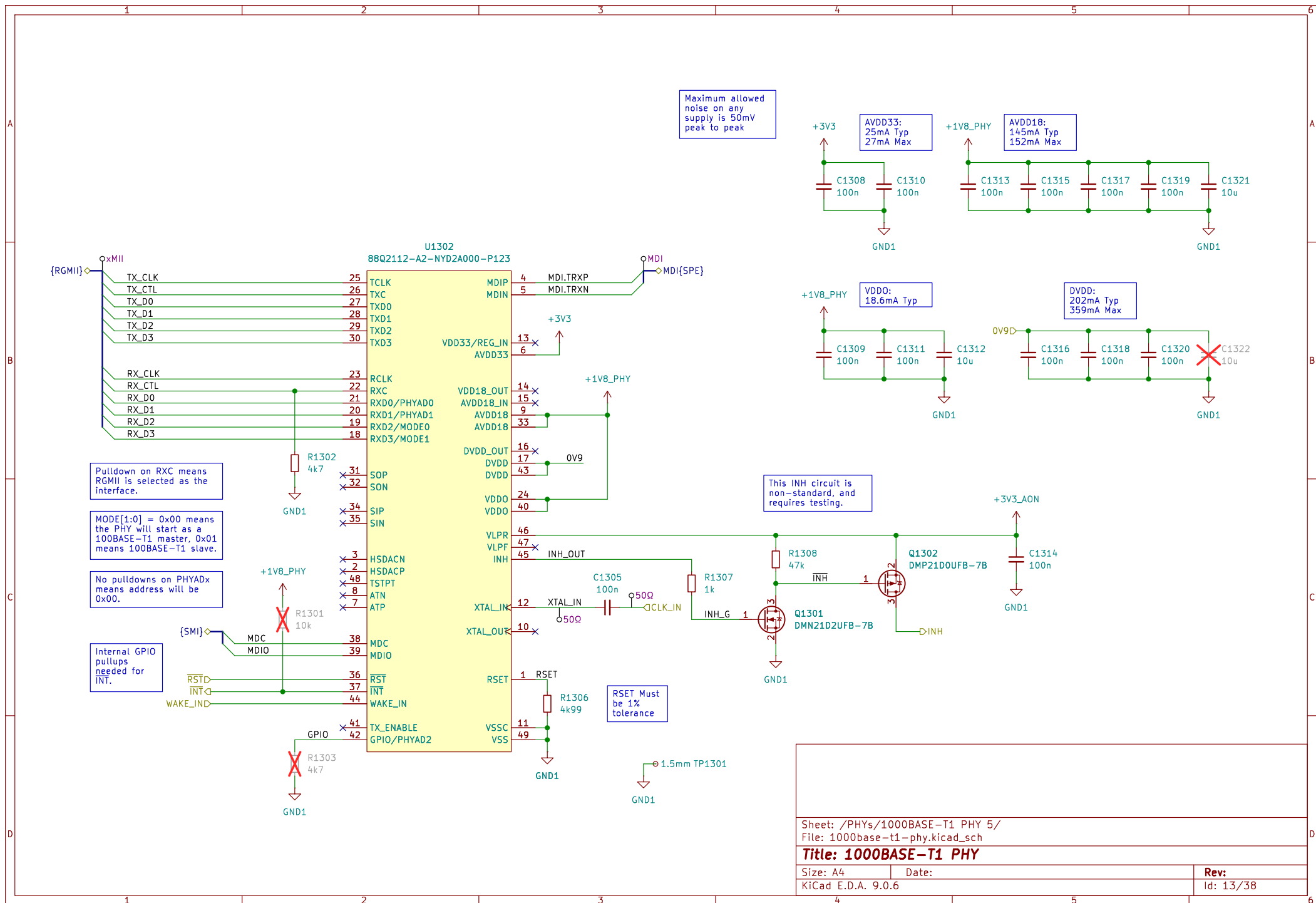
Id: 7/38



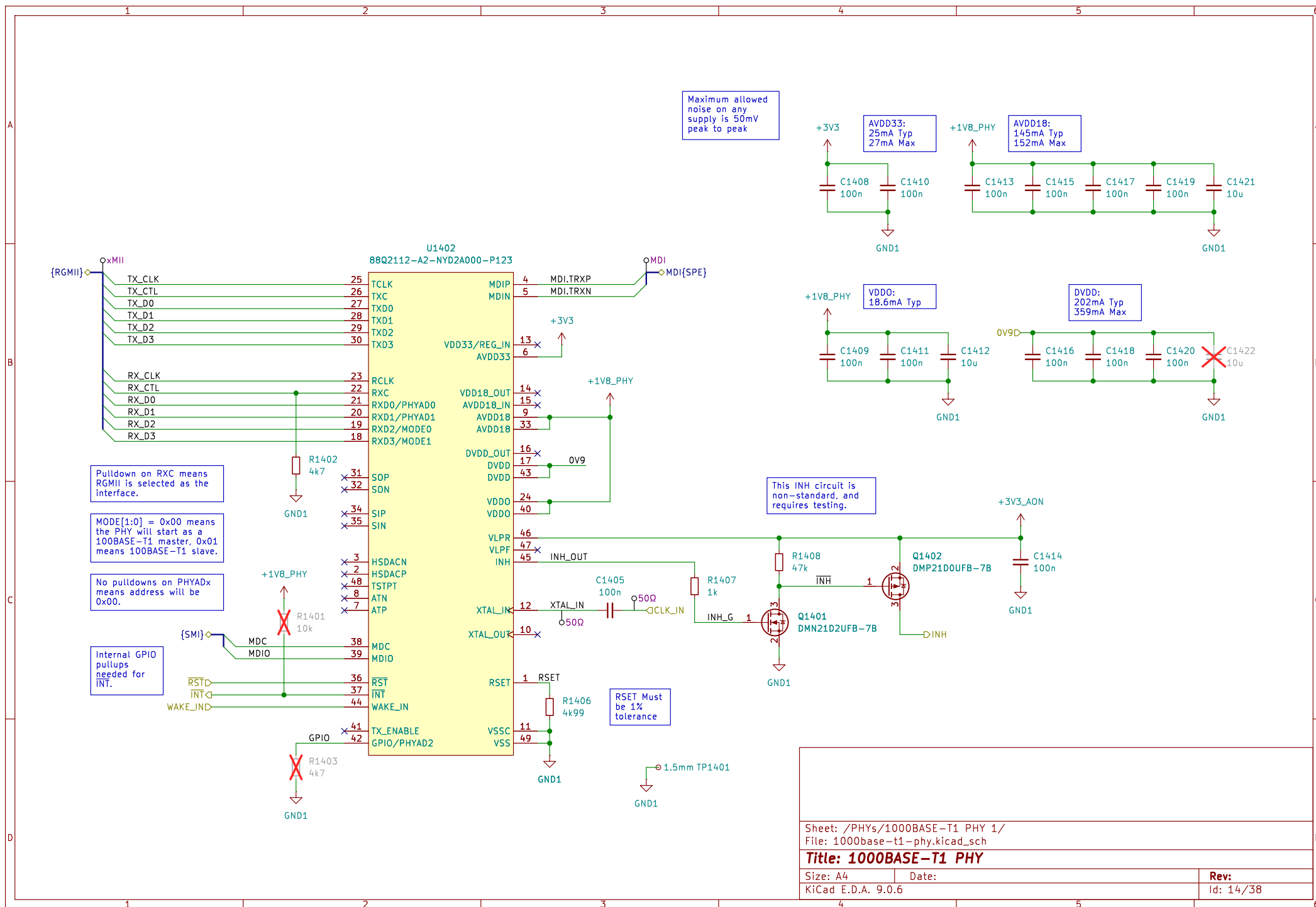
Date:	Rev:
	Id: 8/38

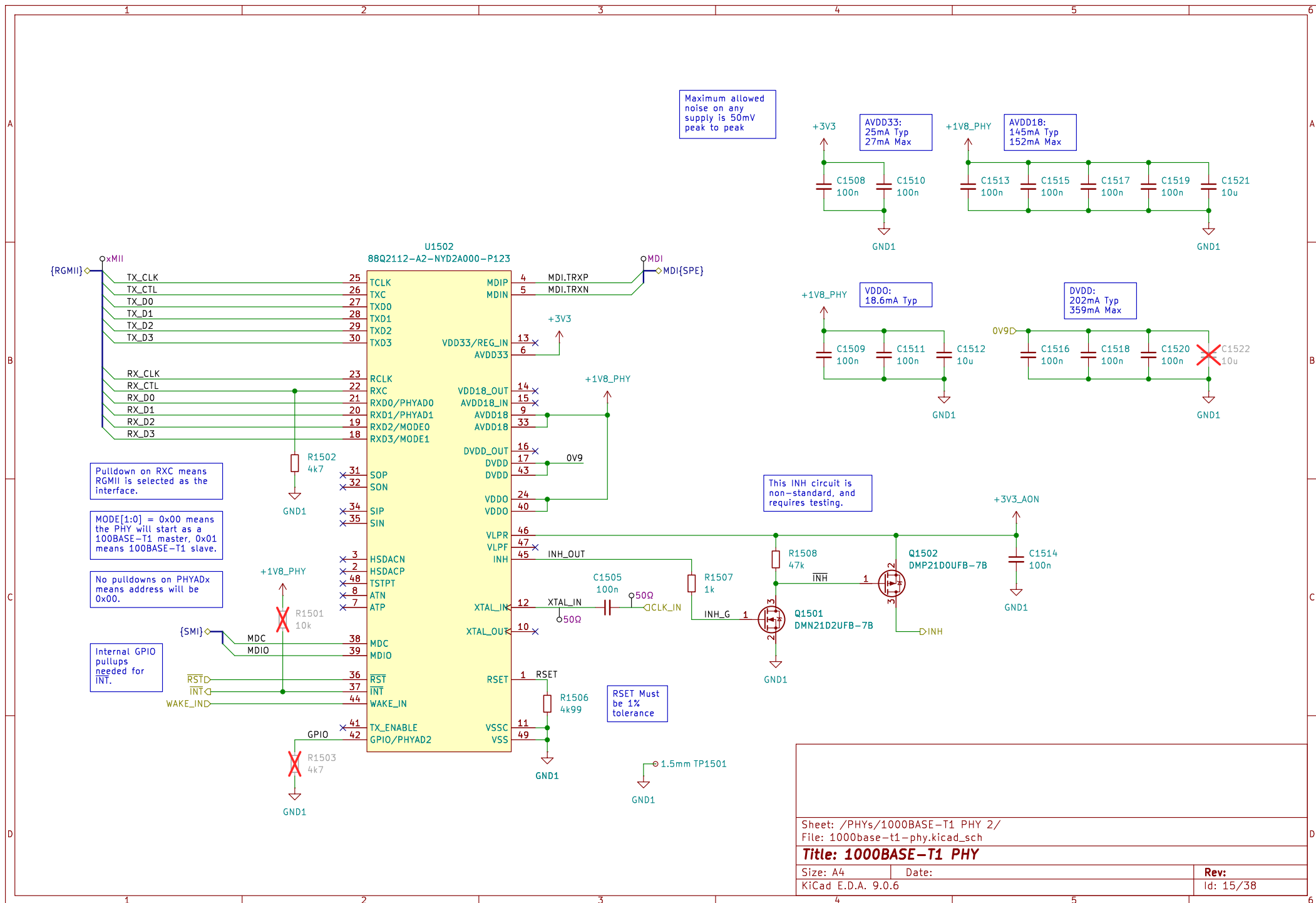


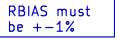




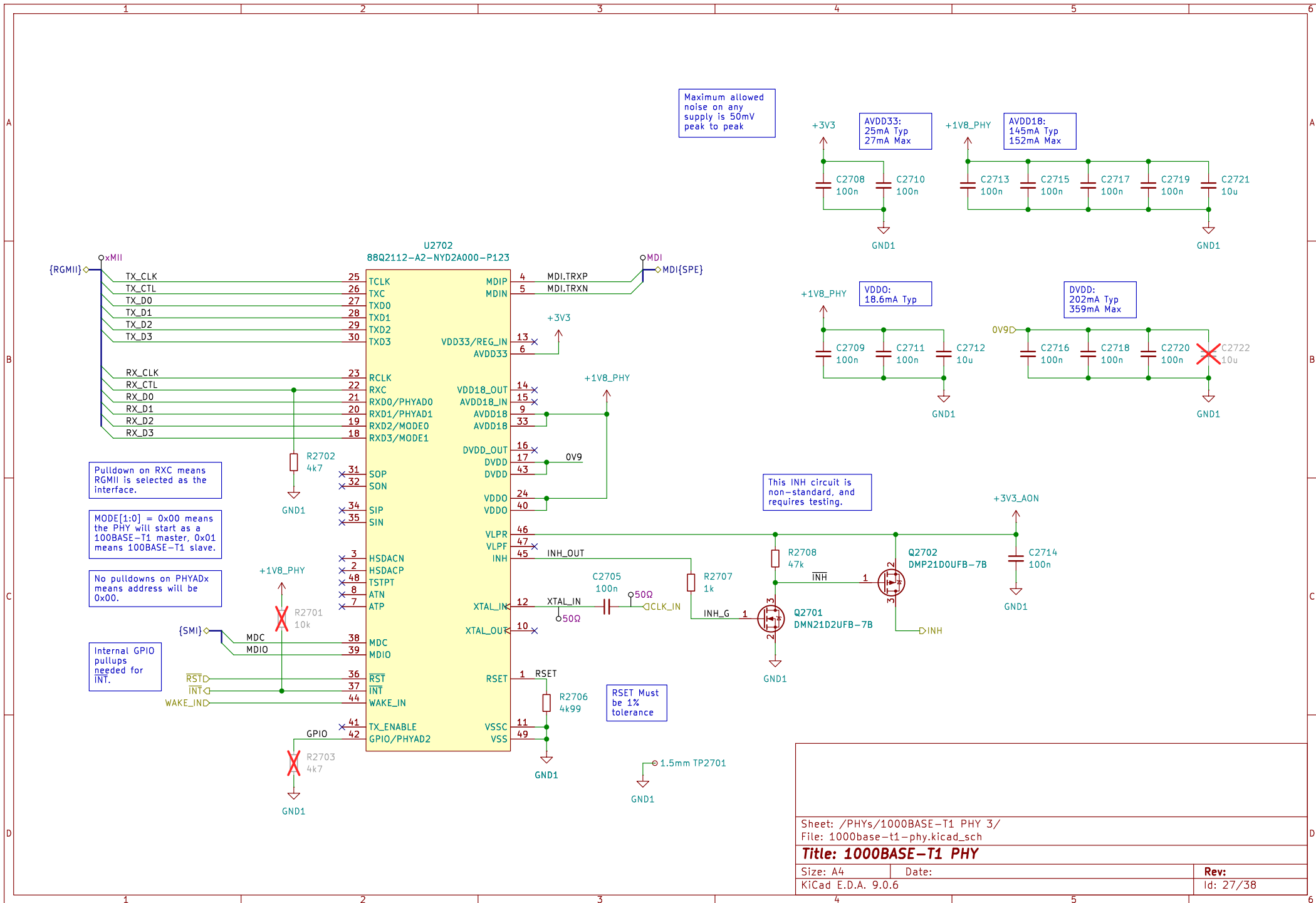
Sheet: /PHYs/1000BASE-T1 PHY 5/	
File: 1000base-t1-phy.kicad_sch	
Title: 1000BASE-T1 PHY	
Size: A4	Date:
KiCad E.D.A. 9.0.6	Rev: Id: 13/38

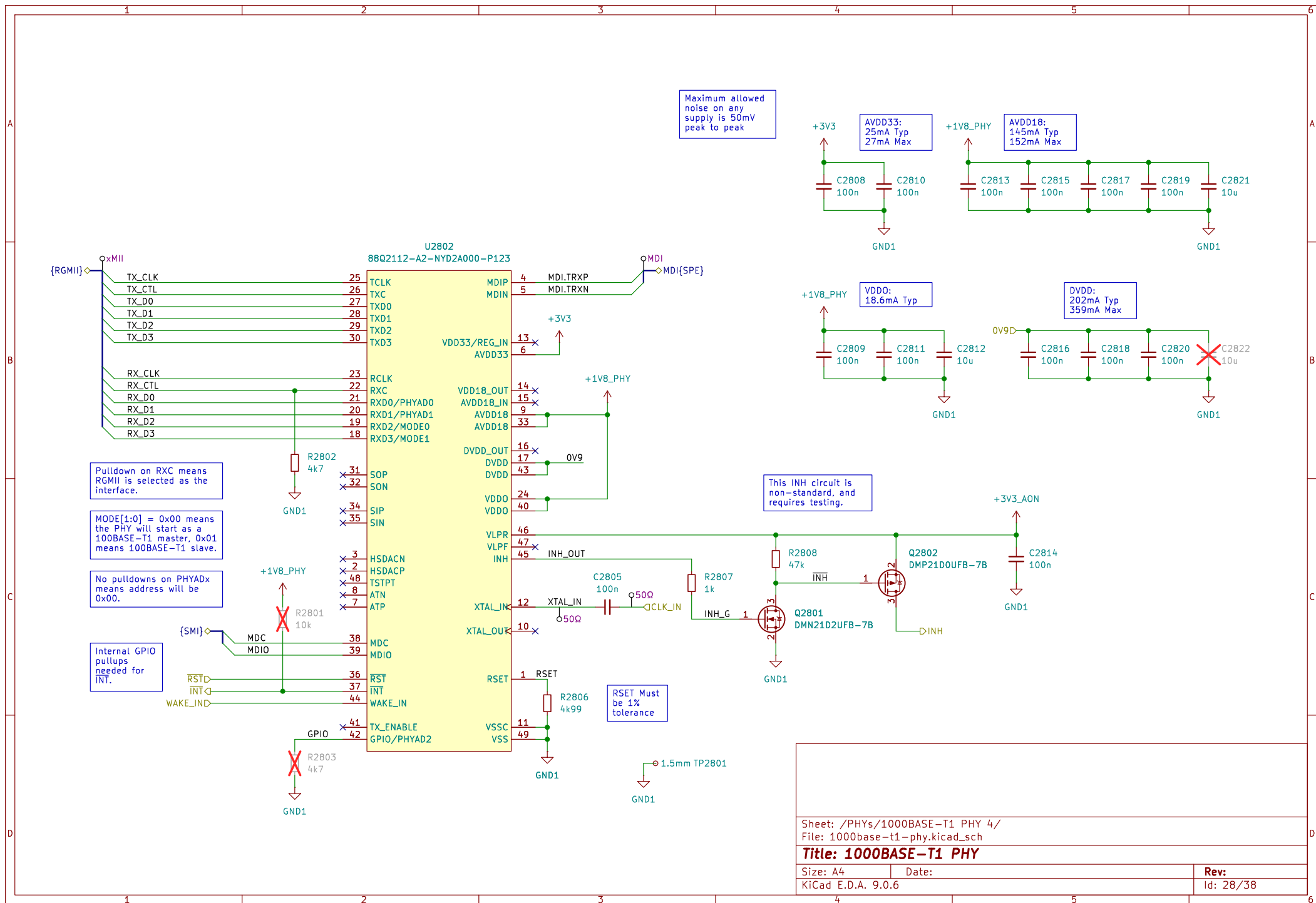


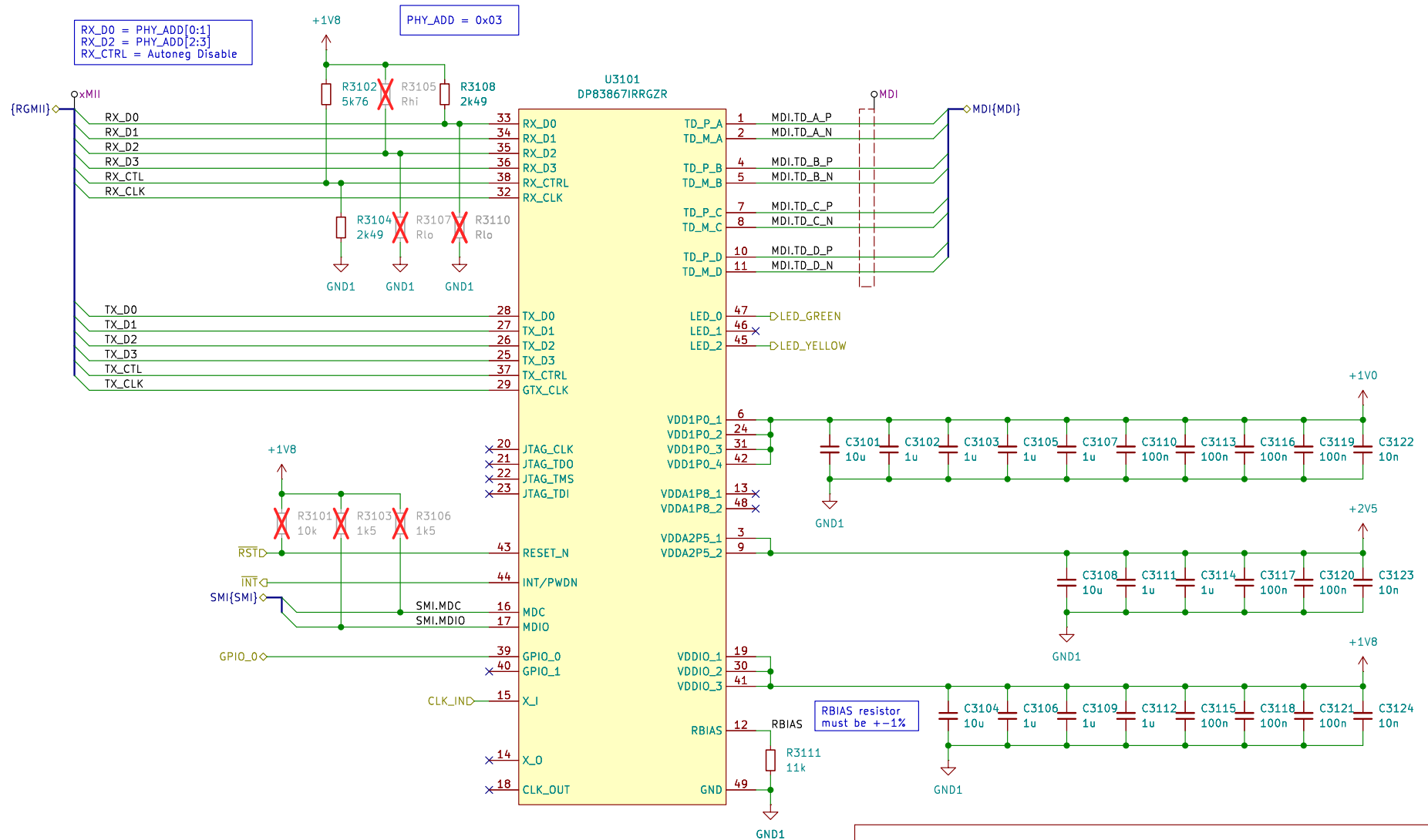




Id: 16/38







Sheet: /PHYs/1000BASE-T PHY/
File: 1000base-t-phy.kicad_sch

Title: PHY

Size: A4
KiCad E.D.A. 9.0.6

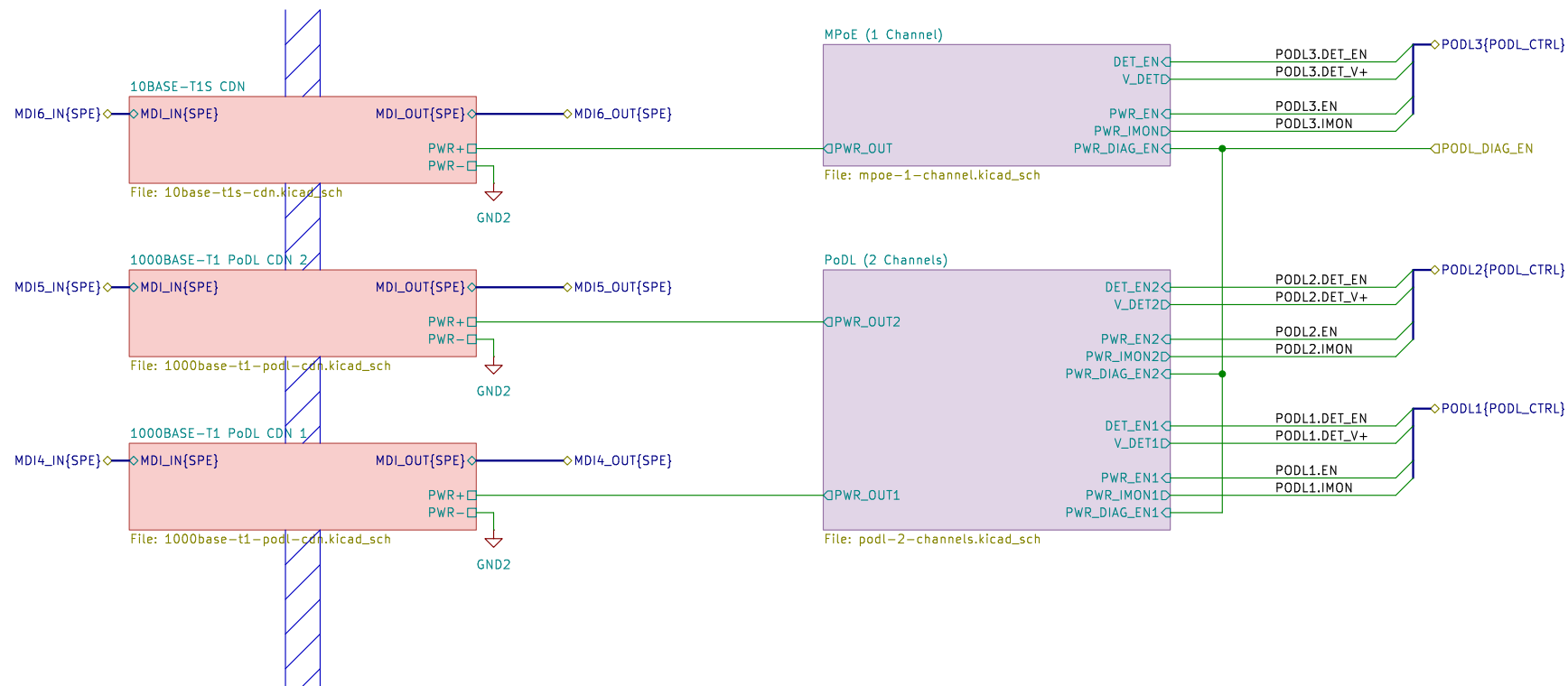
Date:

Rev: 1.0.0

Id: 31/38

Coupling-Decoupling Networks (CDNs)

Power Injection



Coupling-decoupling networks (CDNs) and power over datalines (PoDL).

Sheet: /CDNs & PoDL/
File: cdn-podl.kicad_sch

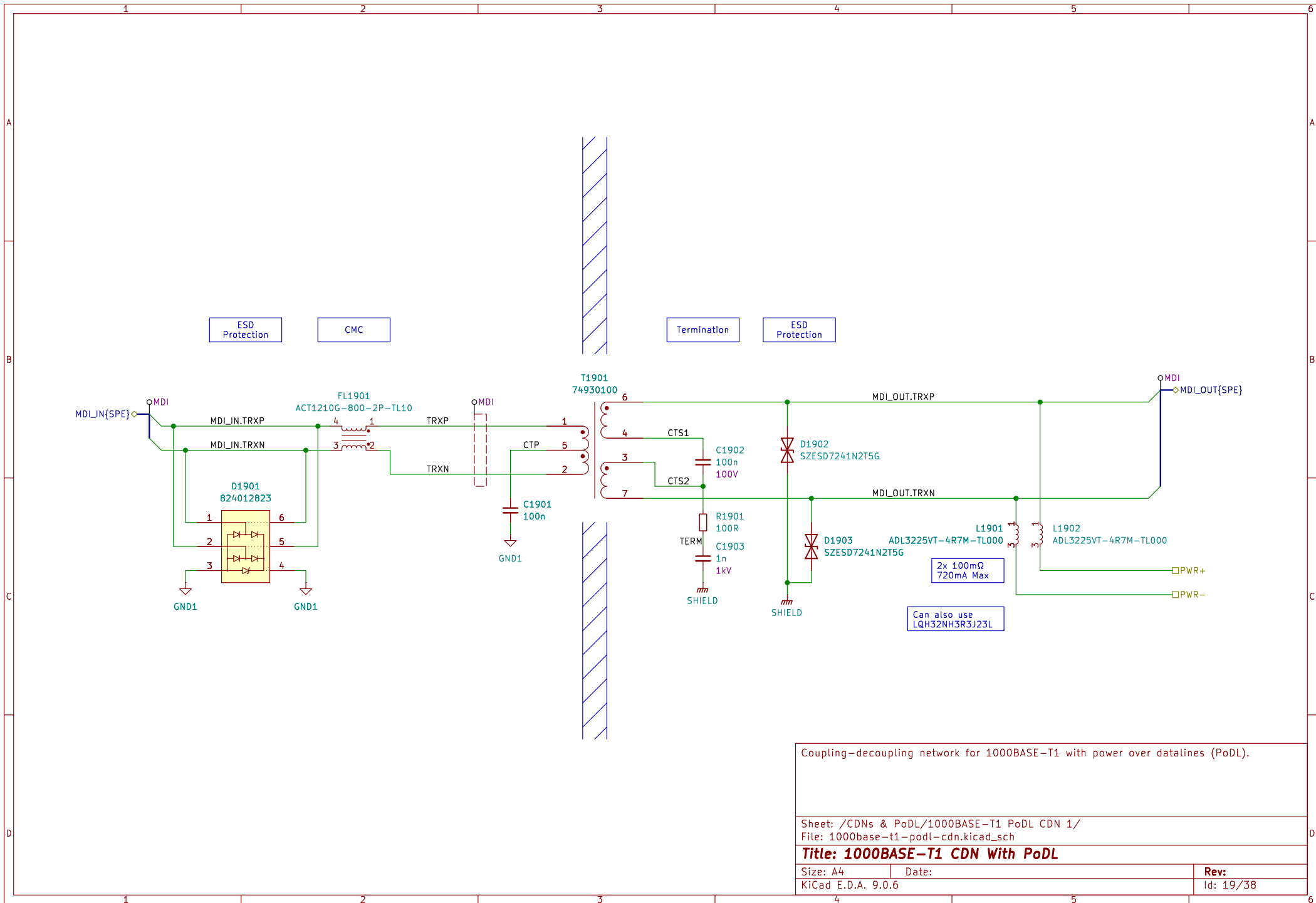
Title: CDNs and PoDL

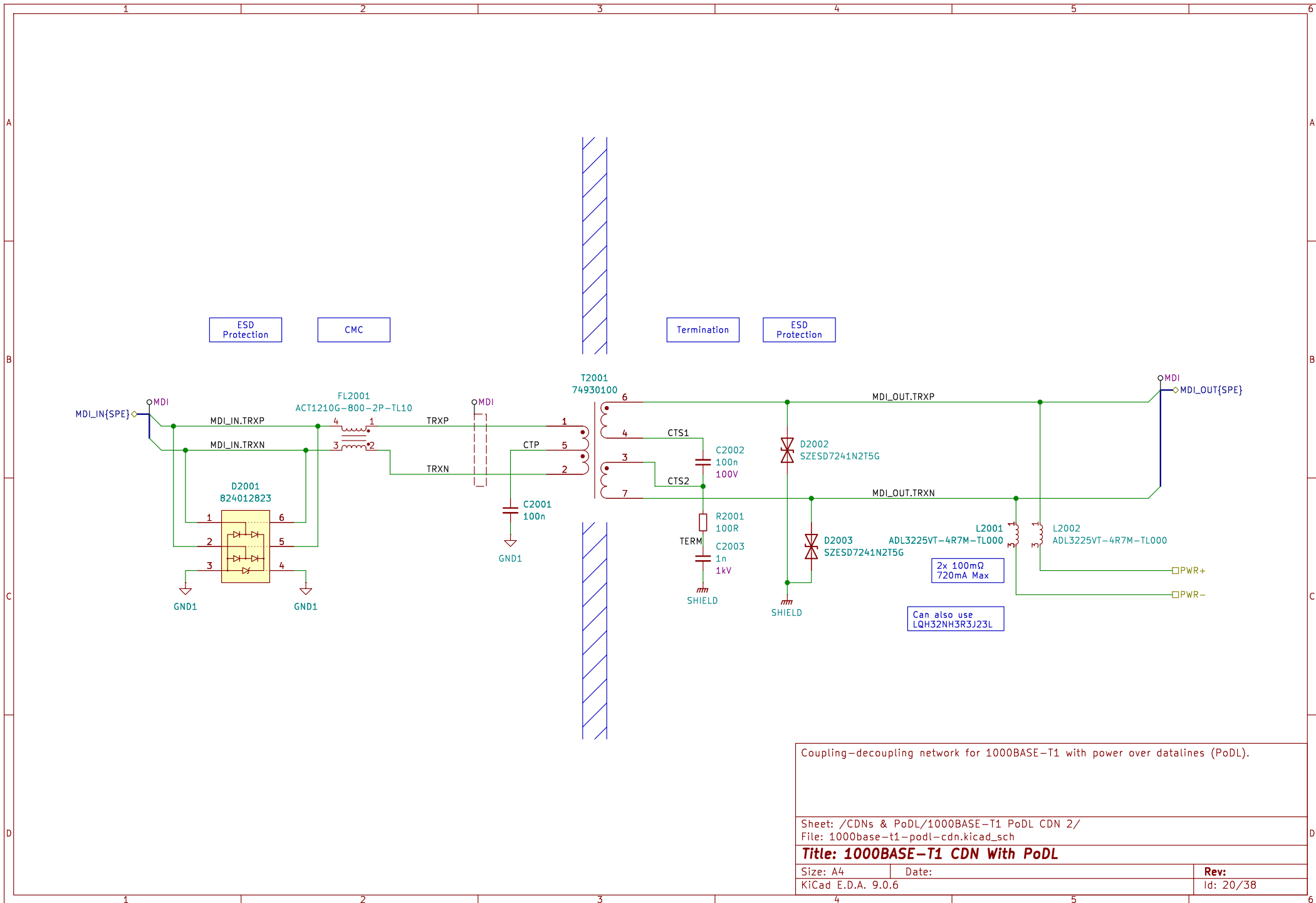
Size: A4
KiCad E.D.A. 9.0.6

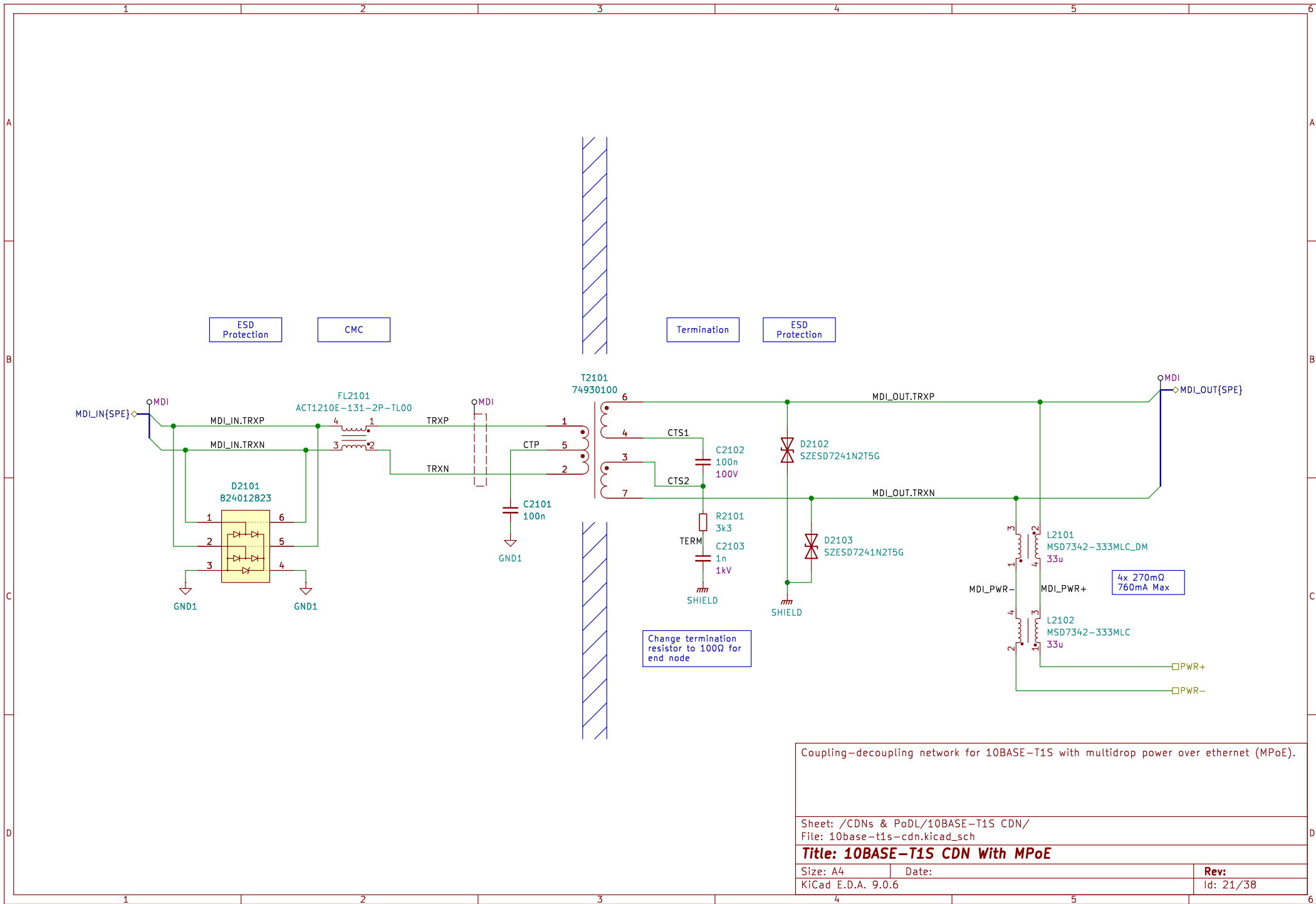
Date:

Rev:

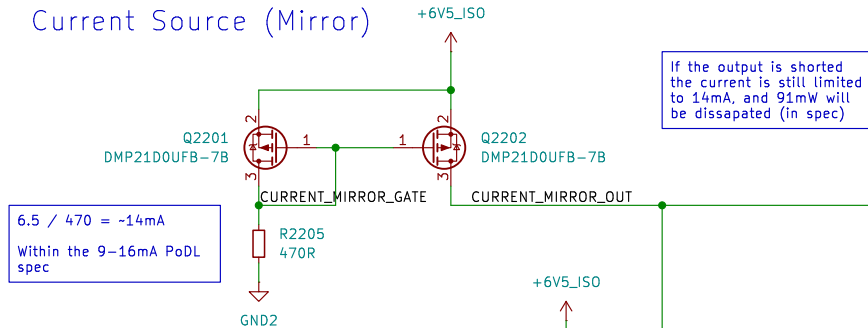
Id: 17/38



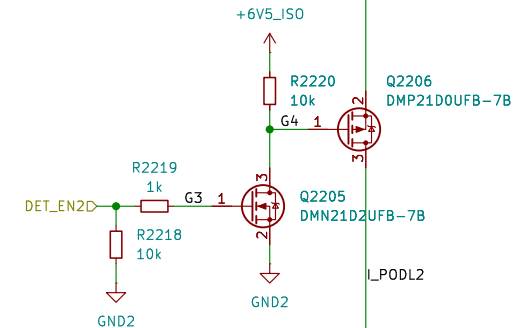
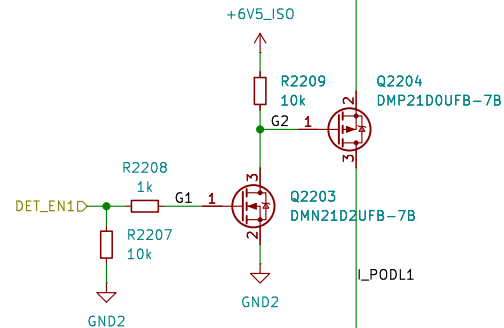




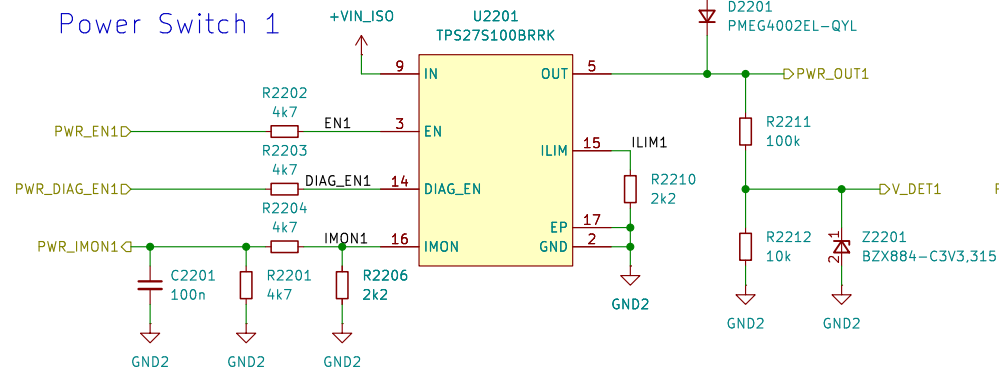
Current Source (Mirror)



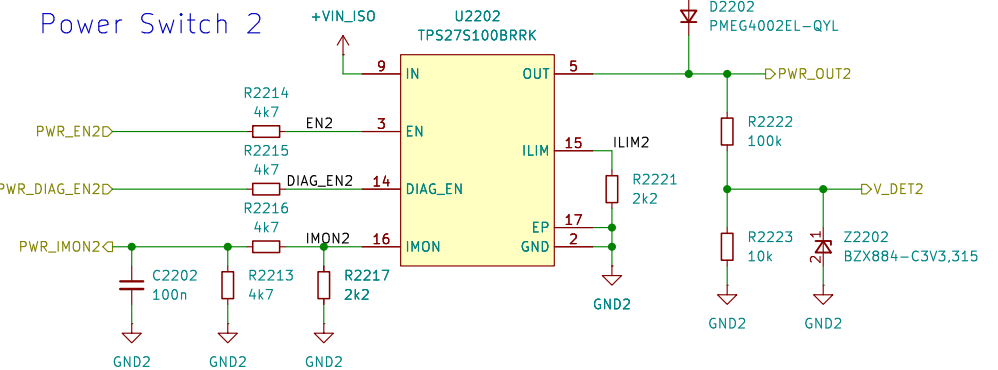
Current Switches



Power Switch 1



Power Switch 2



1.1A Maximum output current
1.1A Maximum measurable current

1/11 Potential divider and zener clamp to protect the 3V3 microcontroller IO.

IMON can go up to 4.75V, potential divider limits this to be within 2.5V ADC range.

$$I_{OUT} = 0.28046 \cdot V_{MON}$$

During the detection phase, 9-16mA is sent into TRXP and the resulting voltage is measured.
If a PoDL compatible device is detected on the other end (indicated by a 4.05-4.55V zener diode) then the full 24V is enabled.

Sheet: /CDNs & PoDL/PoDL (2 Channels)/
File: podl-2-channels.kicad_sch

Title: PoDL Detection, Switching and Monitoring

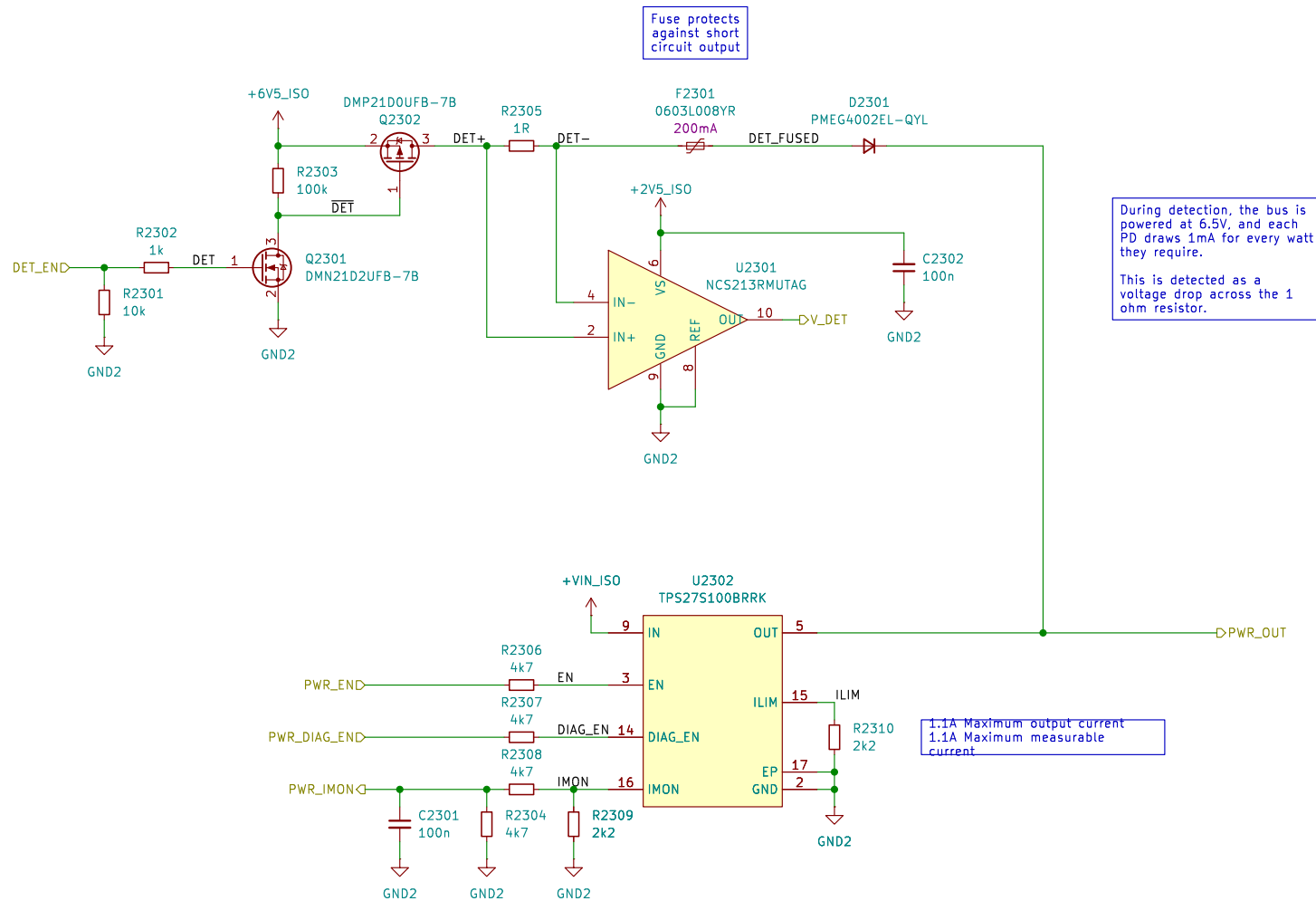
Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.6

Id: 22/38



Fuse protects against short circuit output

During detection, the bus is powered at 6.5V, and each PD draws 1mA for every watt they require.
This is detected as a voltage drop across the 1 ohm resistor.

1.1A Maximum output current
1.1A Maximum measurable current

IMON can go up to 4.75V, potential divider limits this to be withing 2.5V ADC range.

$I_{OUT} = 0.28046 \cdot V_{MON}$

Sheet: /CDNs & PoDL/MPoE (1 Channel)/
File: mpoe-1-channel.kicad_sch

Title: MPoE Detection, Switching and Monitoring

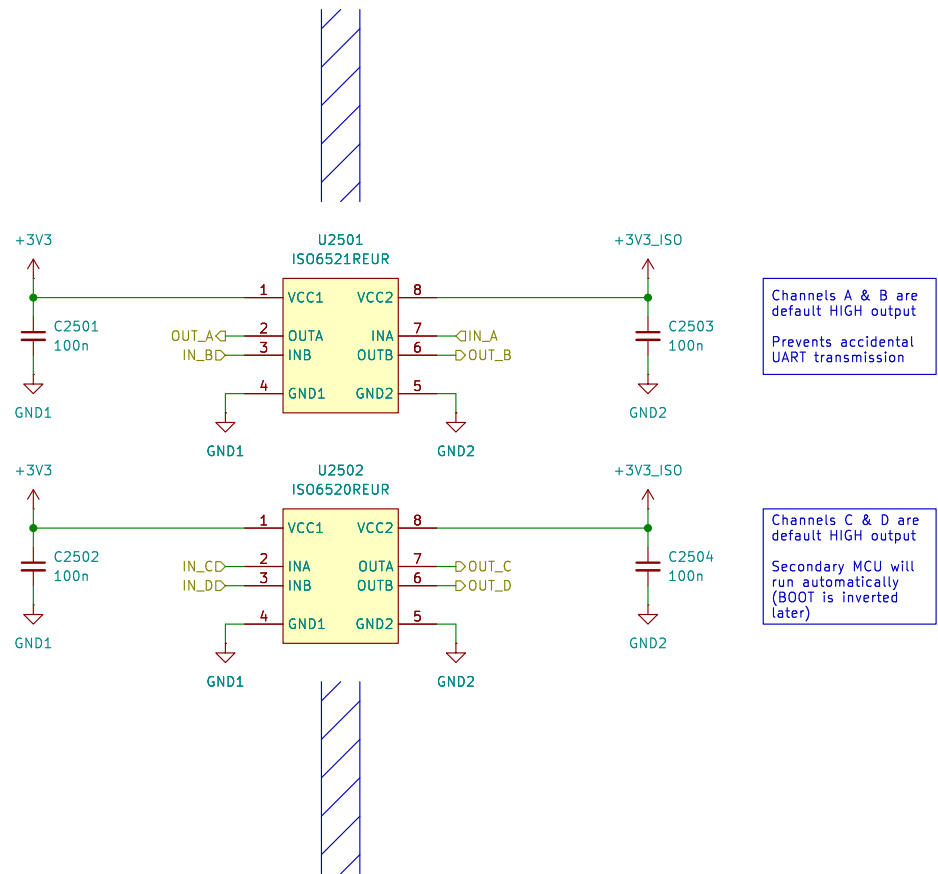
Size: A4

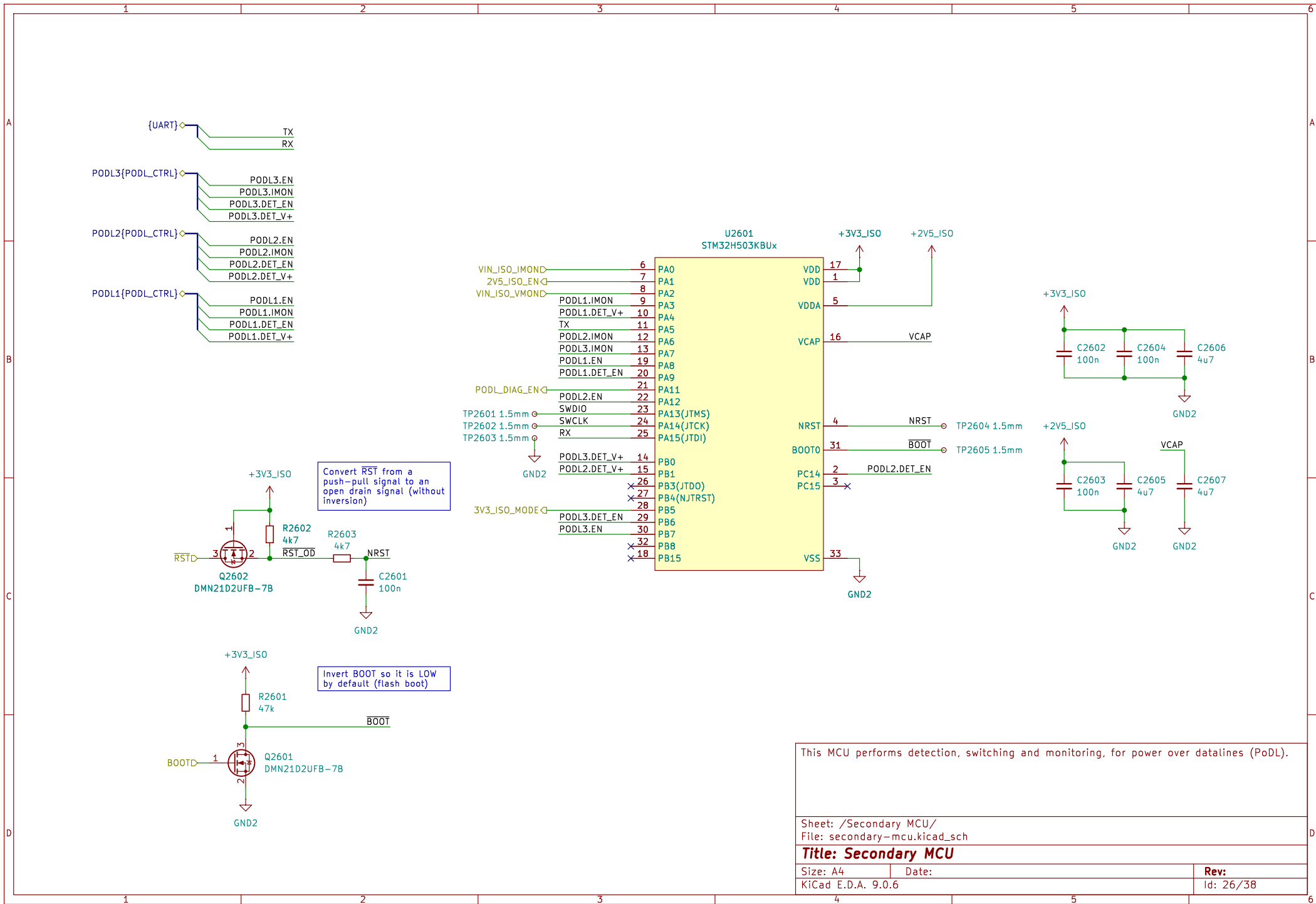
Date:

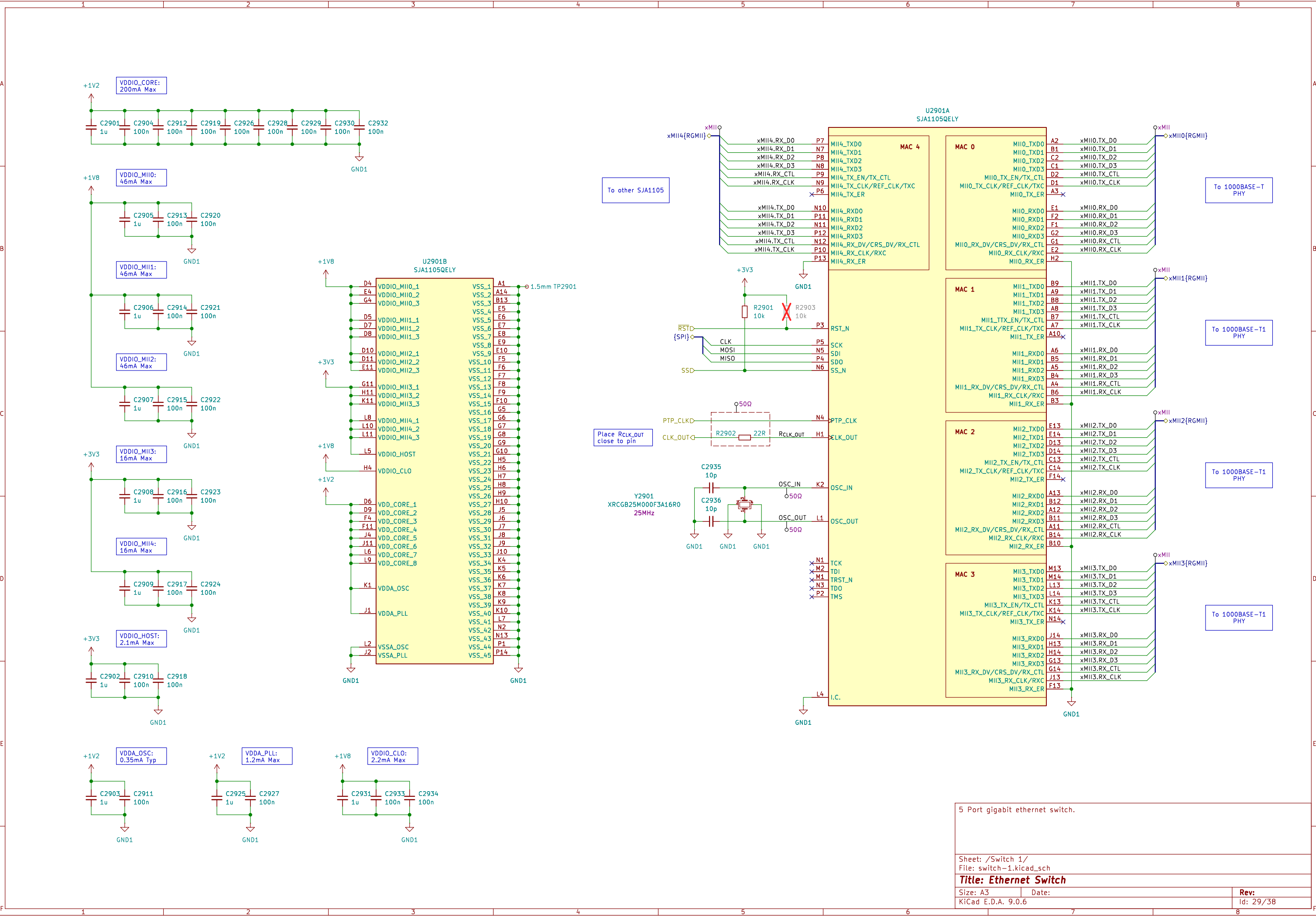
Rev:

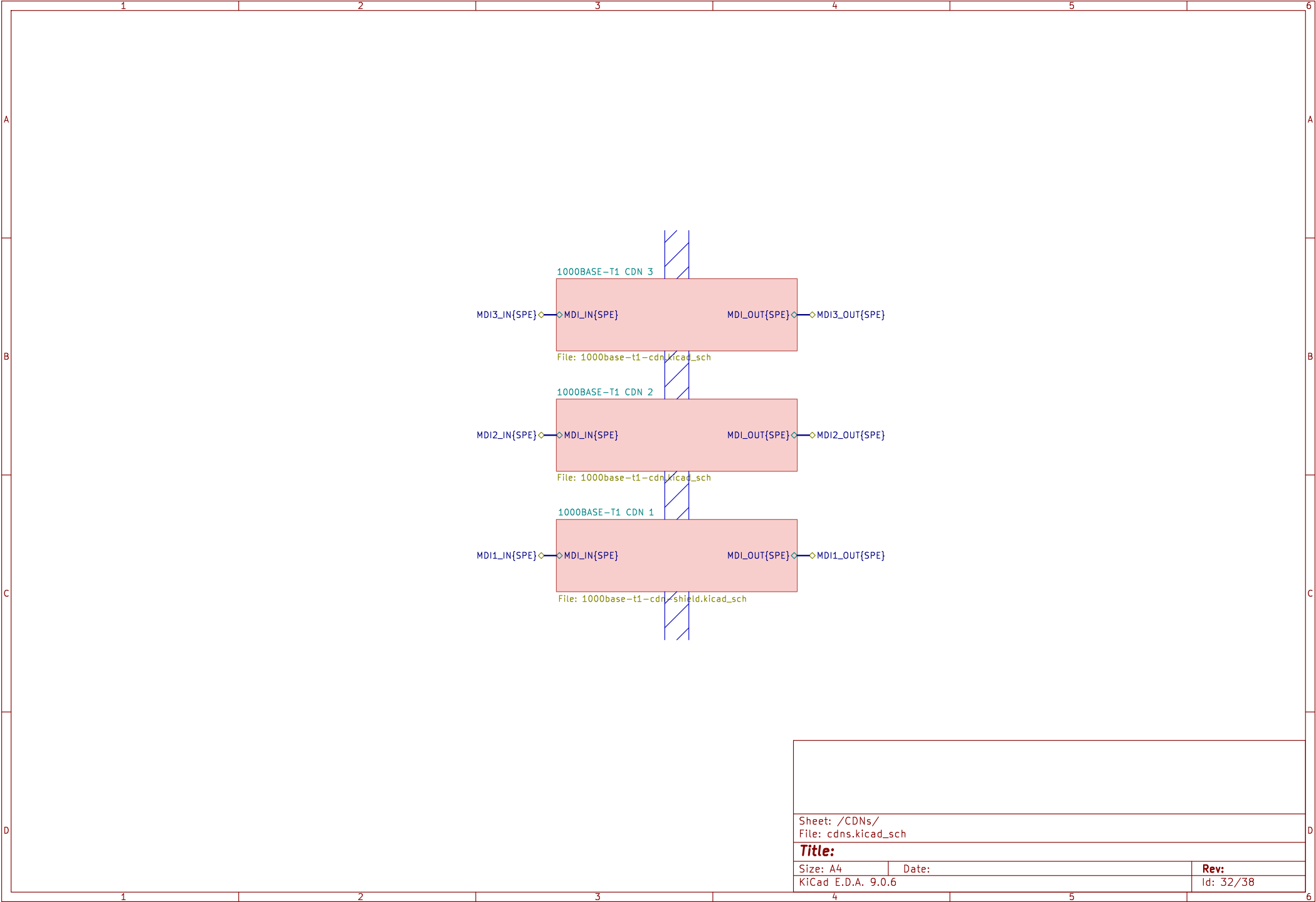
KiCad E.D.A. 9.0.6

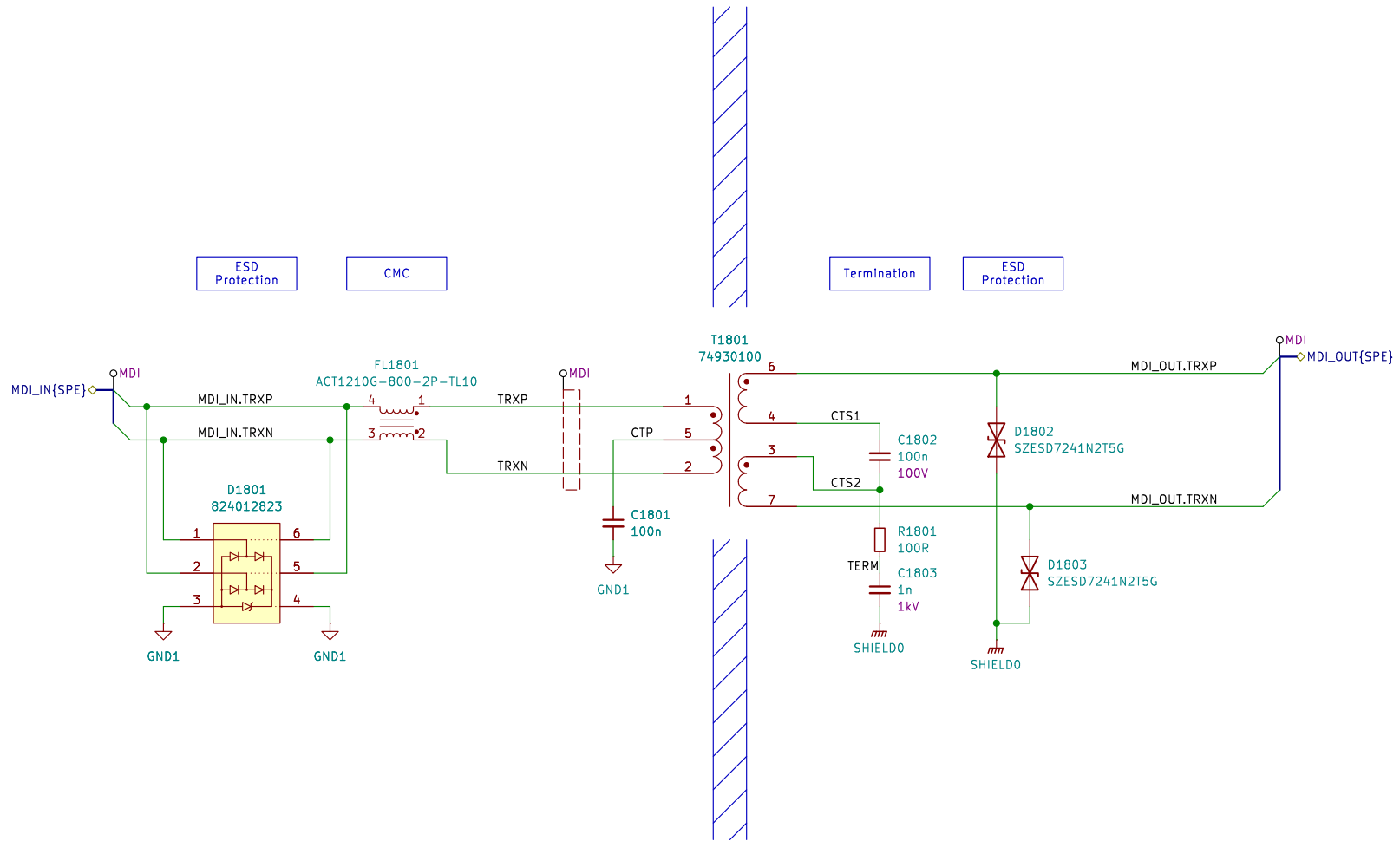
Id: 23/38











Coupling-decoupling network for 1000BASE-T1

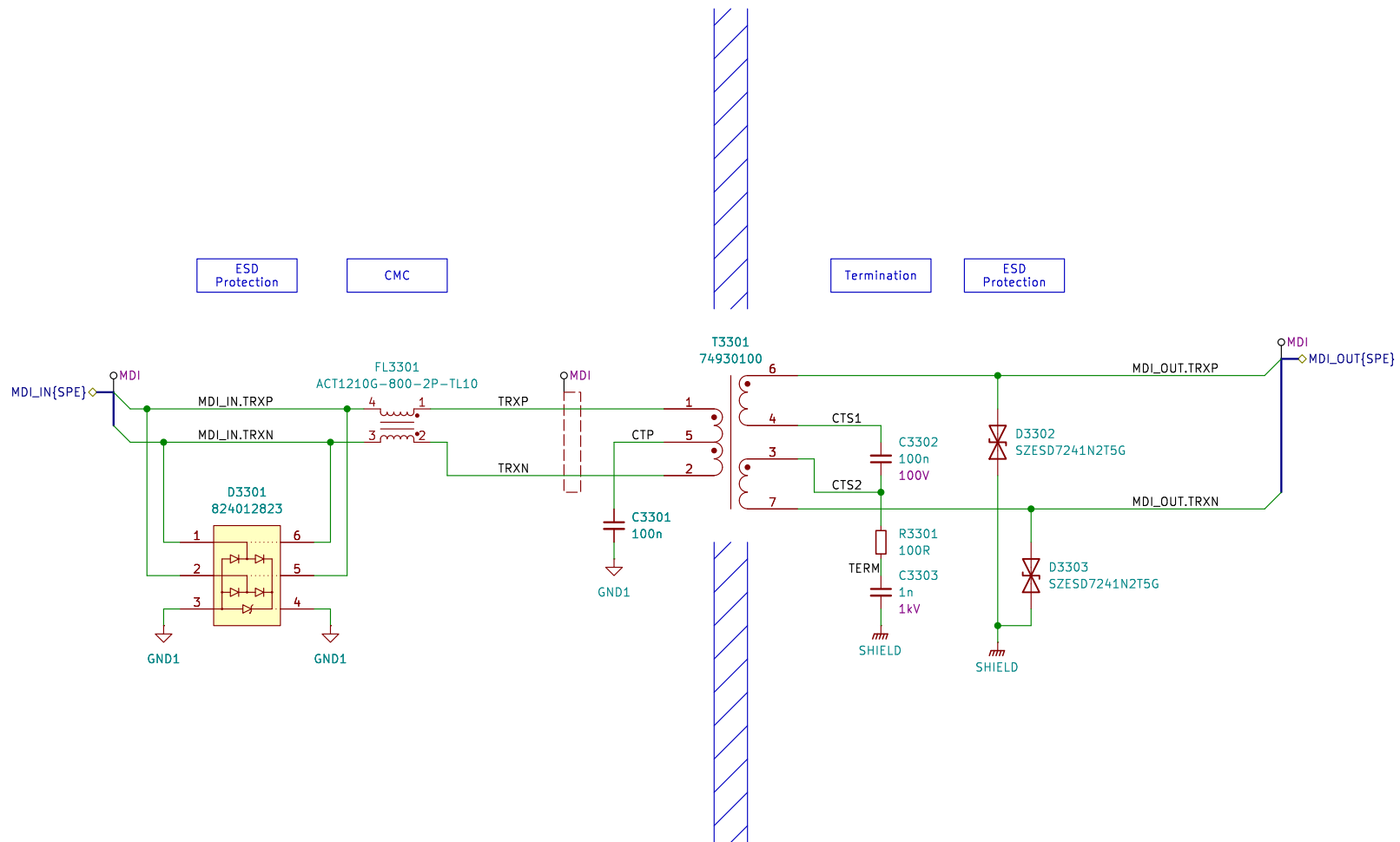
Sheet: /CDNs/1000BASE-T1 CDN 1/
File: 1000base-t1-cdn-shield.kicad_sch

Title: 1000BASE-T1 CDN

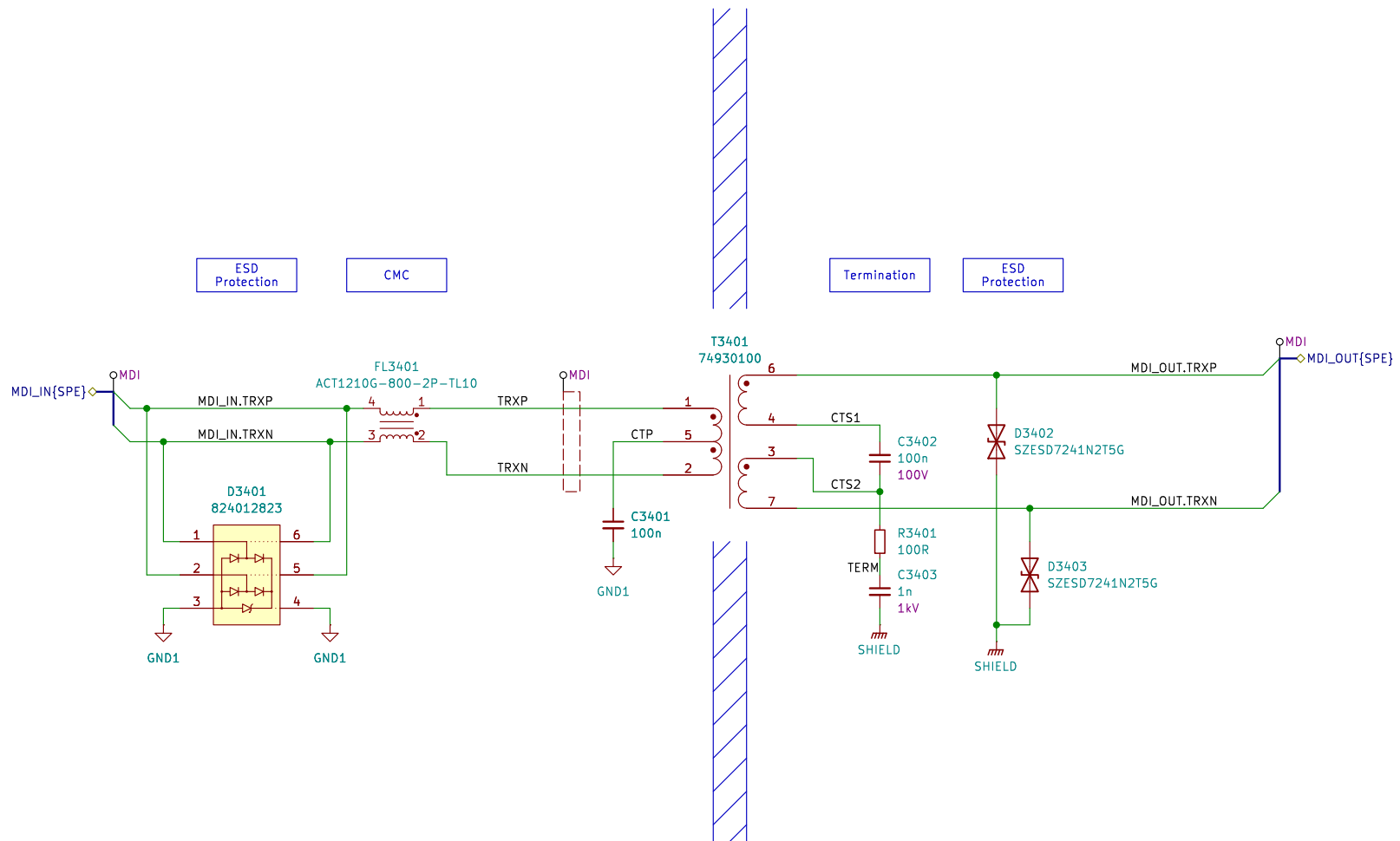
Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 18/38



Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs/1000BASE-T1 CDN 2/ File: 1000base-t1-cdn.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 33/38



Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs/1000BASE-T1 CDN 3/		
File: 1000base-t1-cdn.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 34/38